

External & Internal Anatomy of the Spinal Cord

Principles and Practices of Osteopathic Medicine I

- *Jennifer Xie, Ph.D.*
- Associate Professor
- Biomedical and Anatomical Sciences
- Arkansas, Wilson Hall, Rm # 144J

Email: Jennifer.xie@nyit.edu

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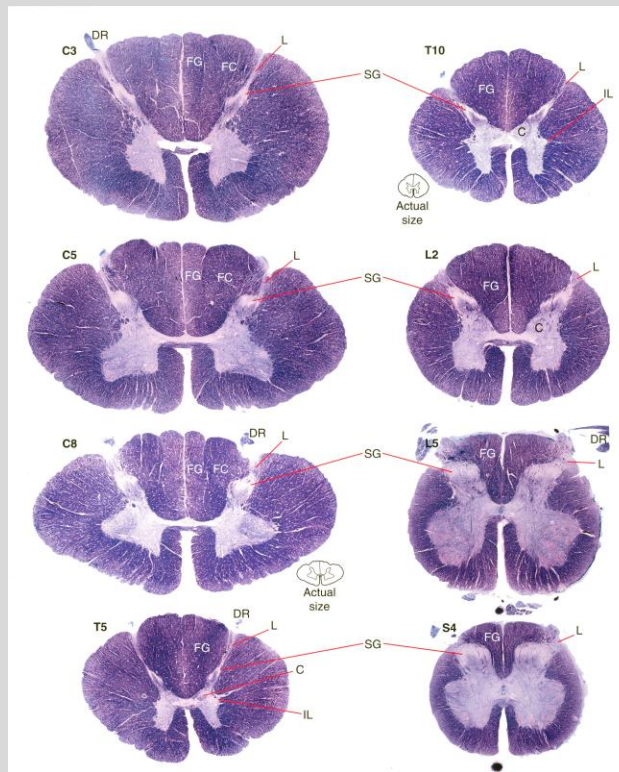
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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

1. Describe the physical geometry of the spinal cord.
2. Identify the major regional **divisions** of the **grey** and **white** matter along with distinctive nuclei and lamina within the grey matter.
3. Describe the pattern, function, and neural basis of **spinal reflexes**.
4. Identify major **ascending tracts** along with their general function.
5. Identify major **descending tracts** along with their general function.
6. Determine the **level of lesions** based on the neurological deficits and abnormalities.



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See **detailed objectives** in CPG
PLEASE READ THE Scholar Rx Brick

Session Objectives Mapped to COMLEX Blueprint

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- **Dimension 2**
 - **Clinical presentation 4.** Patient presentations related to the nervous system and mental health
 - head and spinal cord injury
 - spinal stenosis
 - **Clinical presentation 5.** Patient presentations related to the musculoskeletal system
 - 5.1 Postural abnormalities and spinal deformities
 - 5.2 Back pain and somatic dysfunction of the pelvis, sacrum, and lumbar and thoracic spine
 - 5.3 Neck pain and somatic dysfunction of the cervical spine
 - spinal stenosis, herniated intervertebral discs, spina bifida, spinal fractures, spondylitis, spondyloarthropathies, spondylolisthesis, spondylolysis, spondylosis

Session Objectives Mapped to USMLE Blueprint

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V. Nervous System & Special Senses

A. Normal Processes

2. Organ structure and function

i. Spinal cord

a. gross anatomy and blood supply spinal reflexes

iv. Sensory systems

c. brain and spinal cord (upper moto neuron) basal ganglia and cerebellum

B. Abnormal Processes: Health and Health Maintenance, Screening, Diagnosis, Management, Risks, Prognosis

4. Disorders relating to the spine, spinal cord, and spinal nerve roots

i. cauda equina syndrome;

ii. spinal artery thrombosis/embolus/infarct;

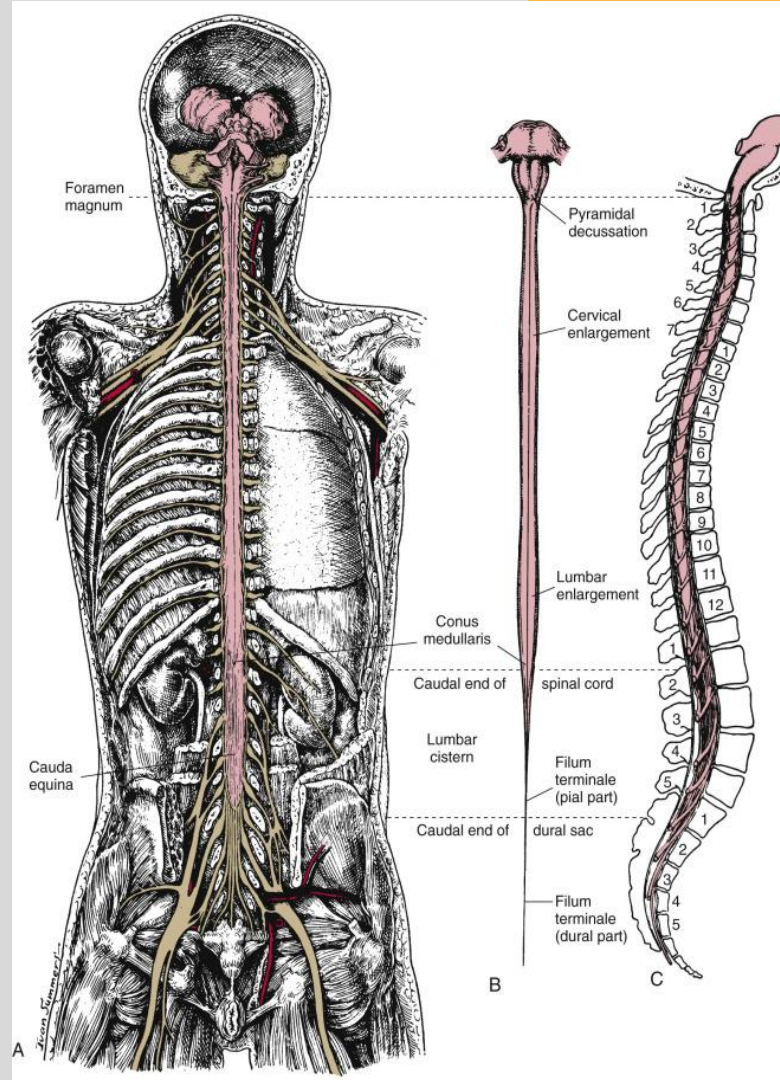
iii. spinal cord compression; spinal cord transection, paraplegia and quadriplegia, acute and chronic effects (eg, autonomic dysreflexia);

iv. spinal stenosis (cervical, lumbar);

v. syringomyelia

Gross anatomy of spinal cord

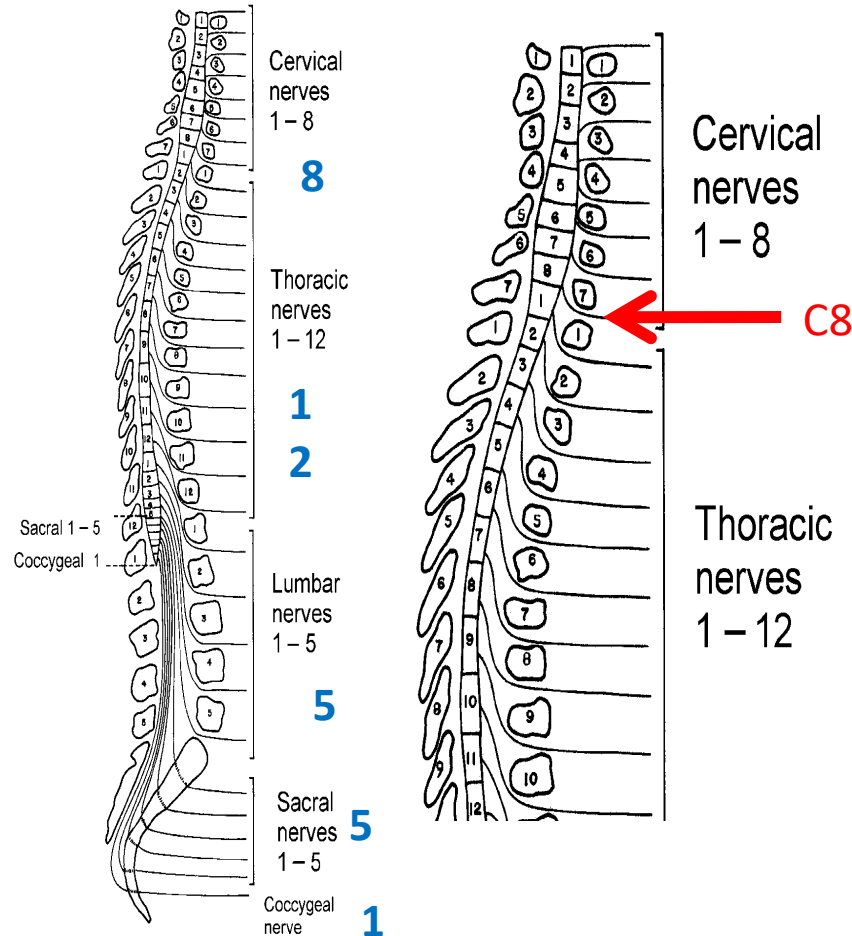
- Cylindrical structure
- Slightly flattened dorsoventrally
- Located in the spinal canal of the vertebral column
- Protection for the cord is provided by:
 - **Vertebrae** and Ligaments
 - 3 layers of **meninges**
 - Cushion of cerebrospinal fluid (**CSF**)



Spinal nerves and spinal vertebrae

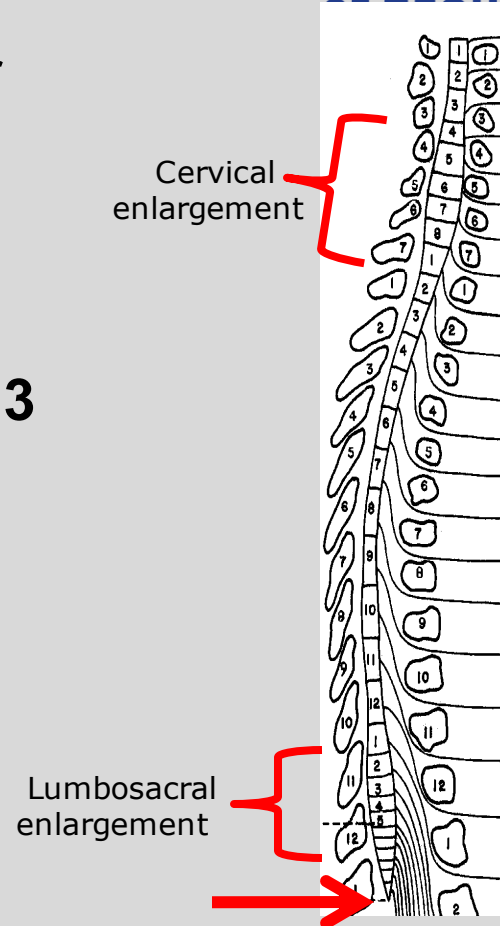
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- **31 pairs** of spinal nerves
- Only **30 vertebrae**
- **C8** spinal nerve exits below C7 vertebra



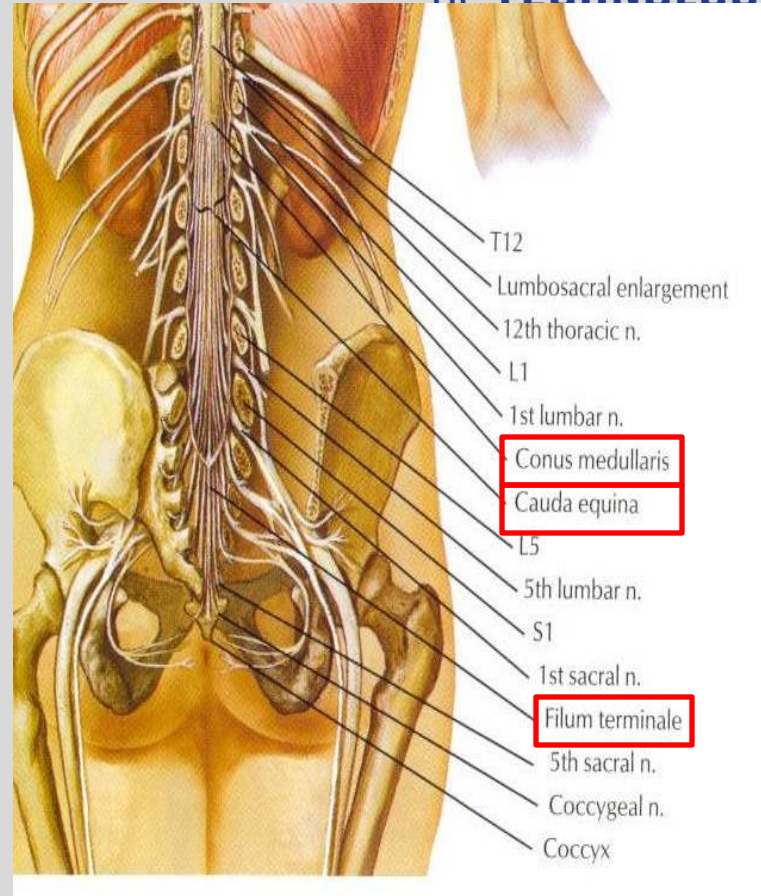
Spinal cord enlargements

- Innervation of the upper and lower extremities
- **Cervical enlargement: C4-T1**
 - spinal nerves of brachial plexus
- **Lumbosacral enlargement: L2-S3**
 - spinal nerves of lumbosacral plexus
- Spinal cord ends around **L1-L2 vertebrae**



Spinal cord development

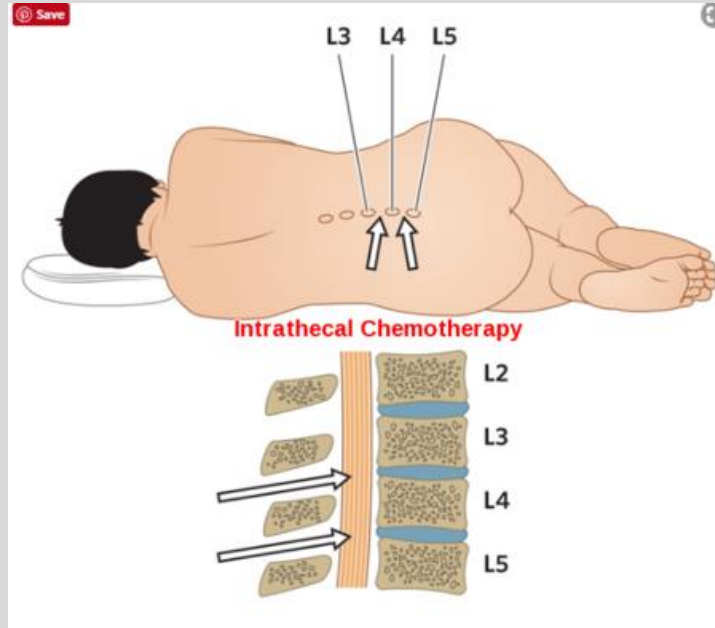
- Until the 3rd month of embryonic life the segments of the spinal cord (neuromeres) correspond with segments of the vertebral column (scleromeres)
- At birth the caudal end of spinal cord is at L2-L3
- By adulthood the caudal end of spinal cord is at **L1-L2**
- **Subarachnoid space** caudal to the end of the spinal cord is the **Lumbar Cistern** which contains roots of lumbar and sacral nerves (**cauda equina**) and cerebral spinal fluid (CSF)



Lumbar puncture

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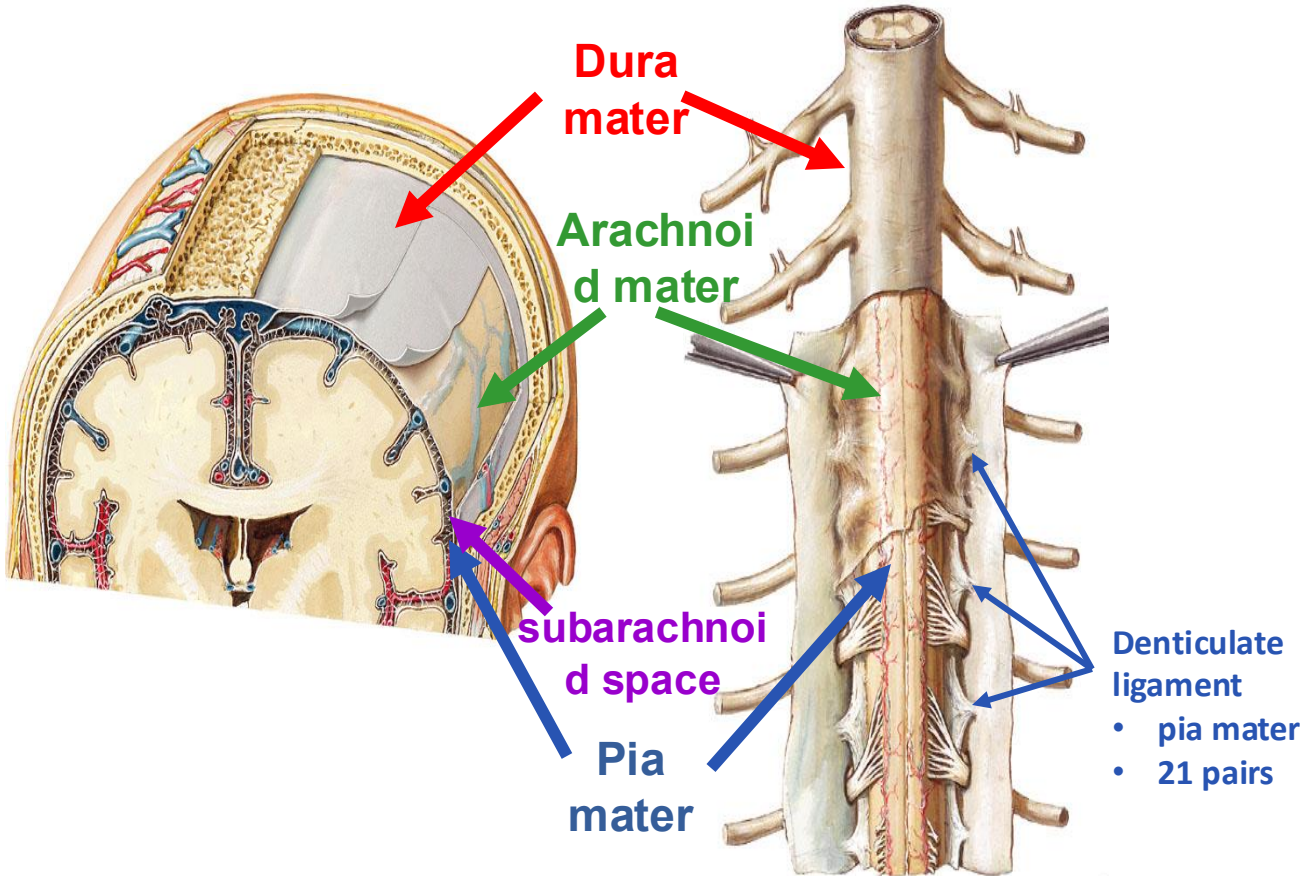
How to

- sample CSF?
- make intrathecal injections?
- make epidural injections?

Meninges of the nervous system

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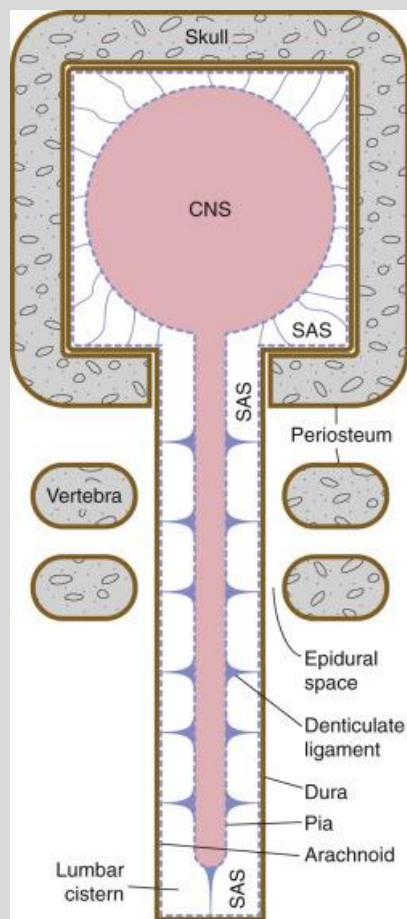
Meninges of the Spinal Cord

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Comparison of cerebrum and spinal cord meninges:

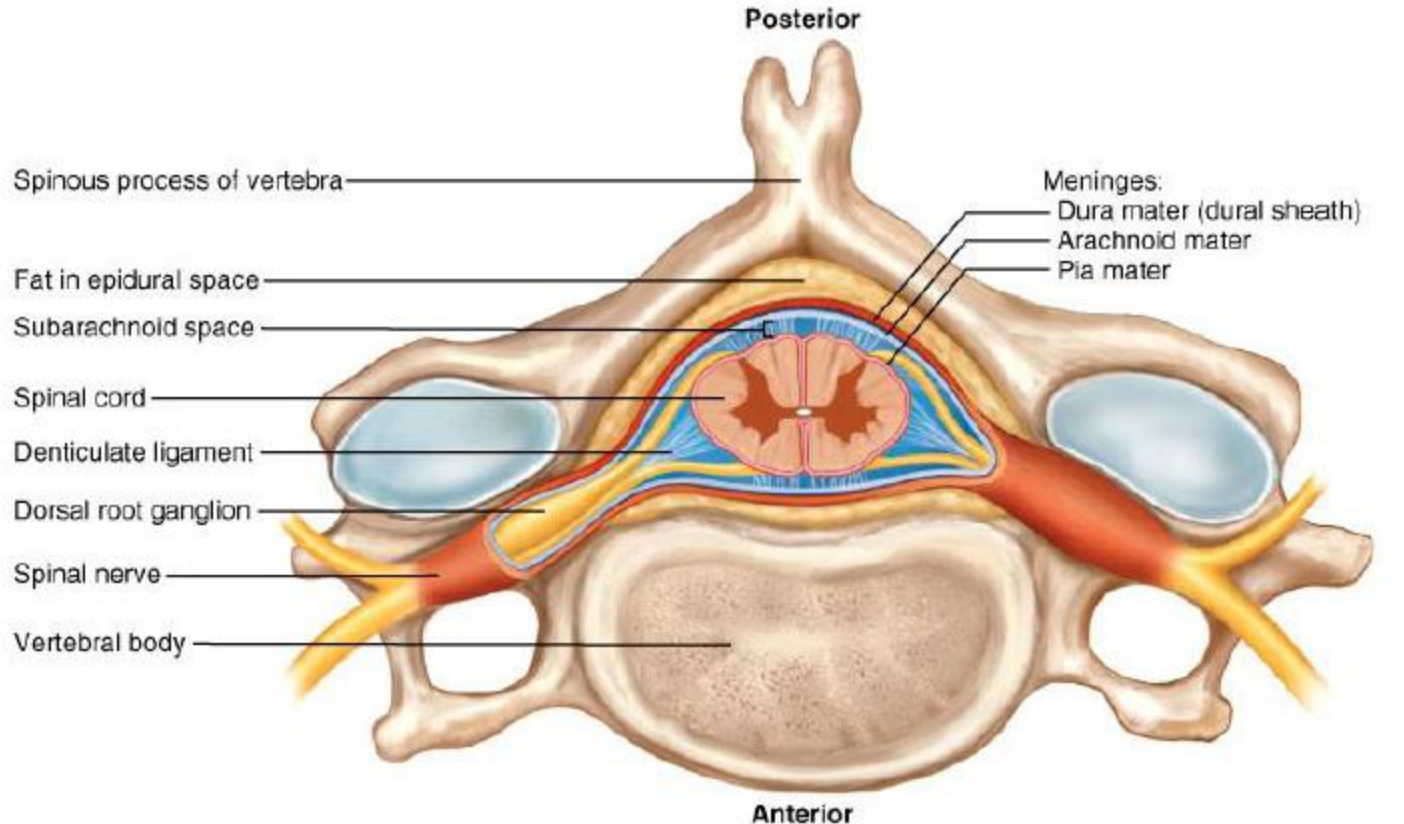
	Cranial	Spinal
Dura mater	Double layered Attached to the inner calvarial surface	Single layered Suspended in vertebral canal
Epidural space	Potential space between periosteum and calvaria	Real space between dura and vertebral periosteum
Arachnoid	Attached to inner surface of dura	Same
Pia mater	Attached to CNS surface	Attached to CNS surface, expanded as denticulate ligaments



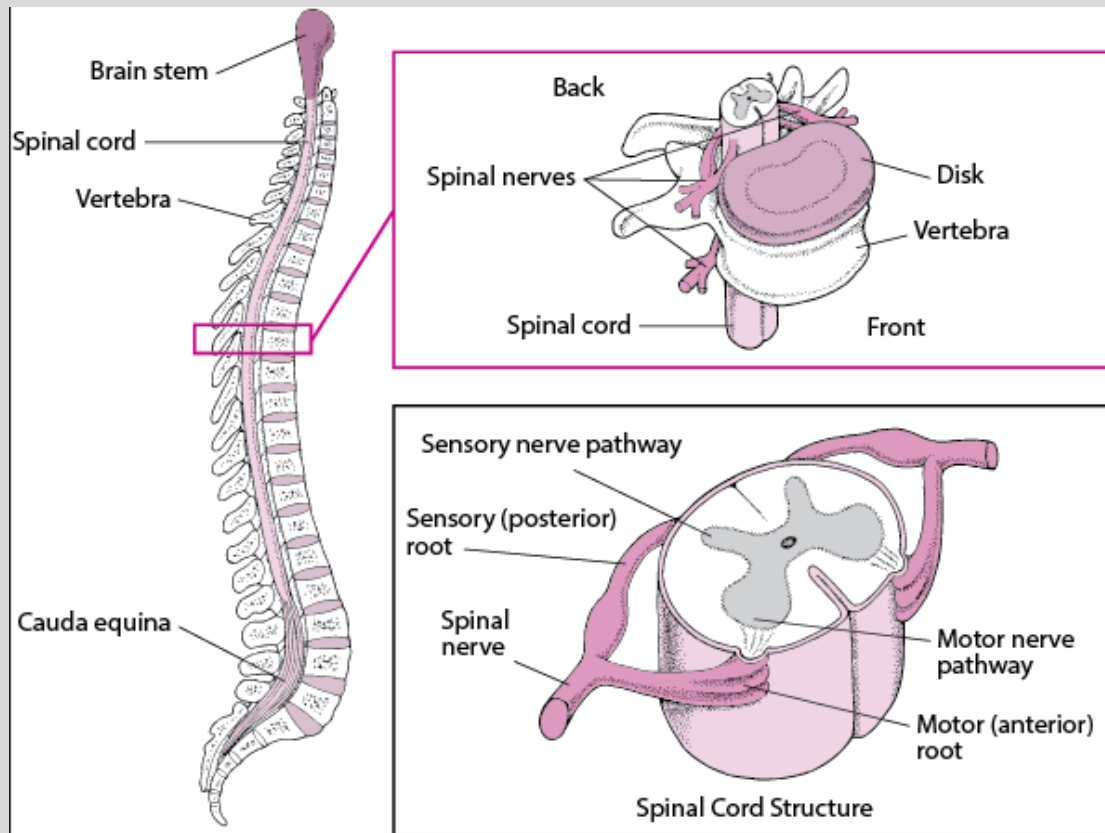
Cross Section Organization of the Spinal Cord

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Cross Section of the Spinal Cord



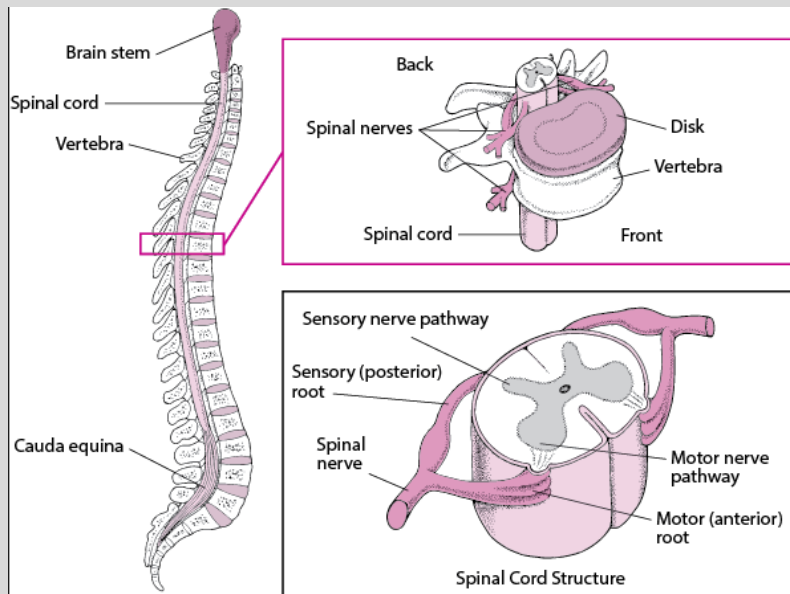
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What happens with a “disc herniation”?

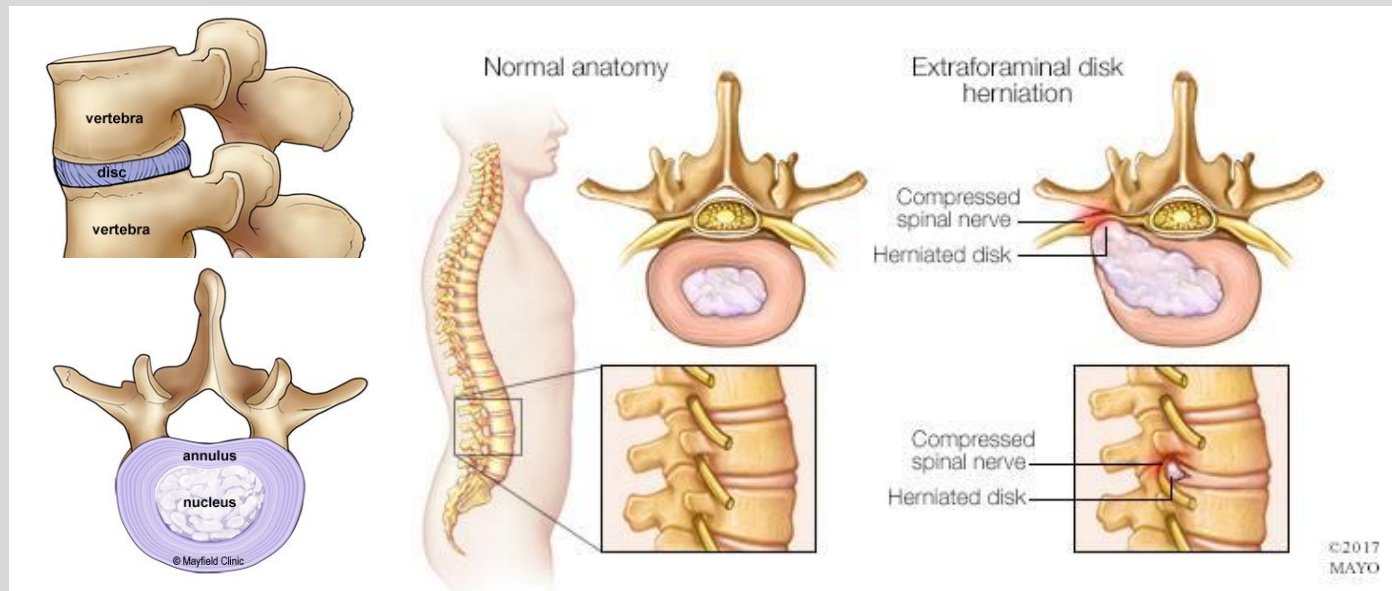
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Lumbar Disk Herniation

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Disc (purple color):

Annulus: a tough fibrous outer ring

Nucleus: gel-filled center

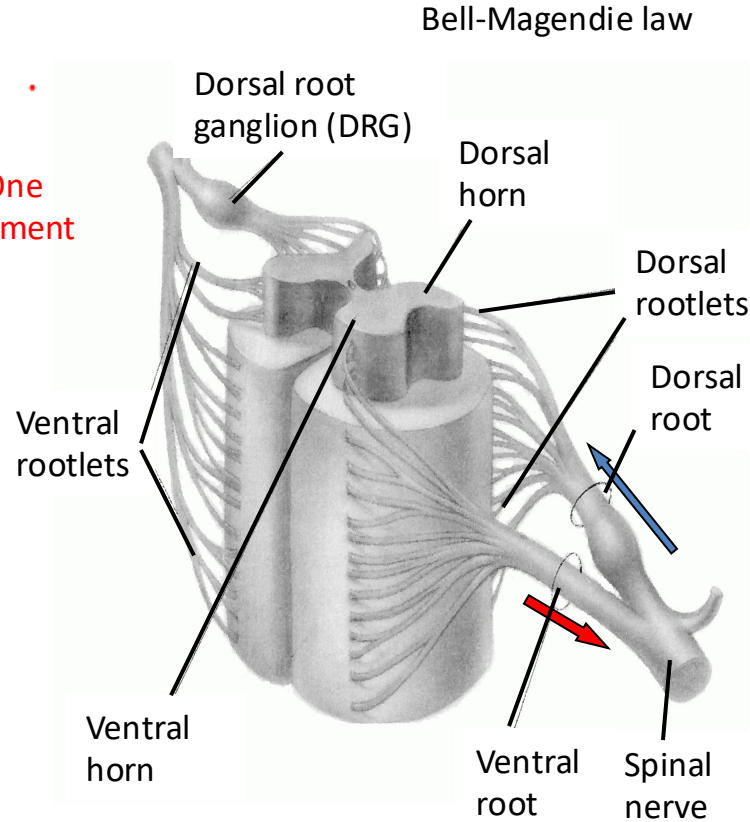
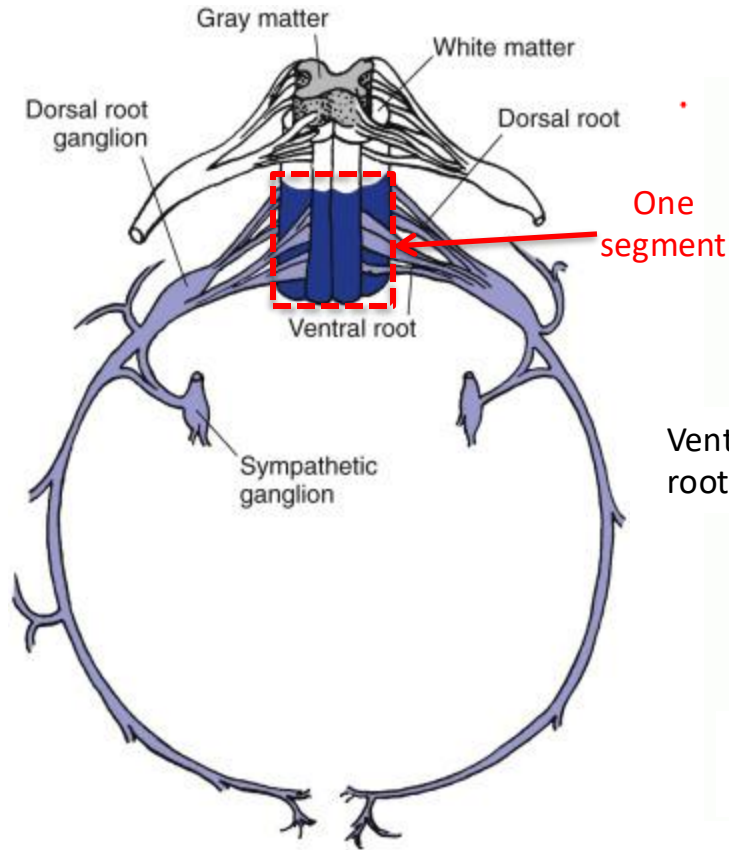
A video:

<https://www.spine-health.com/video/video-can-herniated-discs-heal-their-own>

Cross Section of the Spinal Cord

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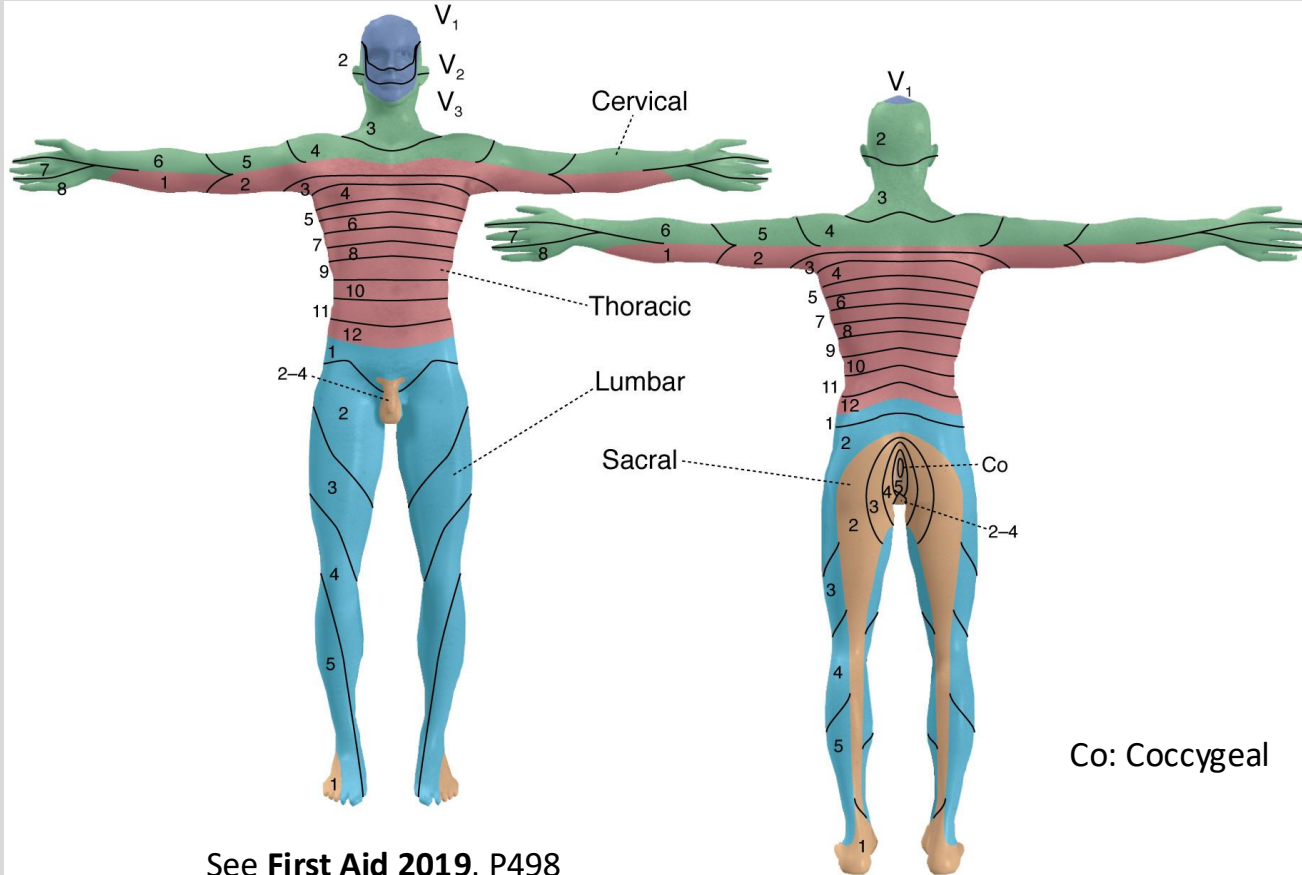
Each segment innervates a **dermatome**.

Dermatomes

- Distribution of cutaneous areas supplied by spinal nerves
- Impairment or loss of cutaneous sensation important in neurological exam

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Co: Coccygeal

See **First Aid 2019**, P498

Shingles



Varicella-zoster virus
(**Herpes**) infection

Same as childhood chickenpox

Becomes dormant in dorsal
root ganglia (**DRG**)

When reactivated –

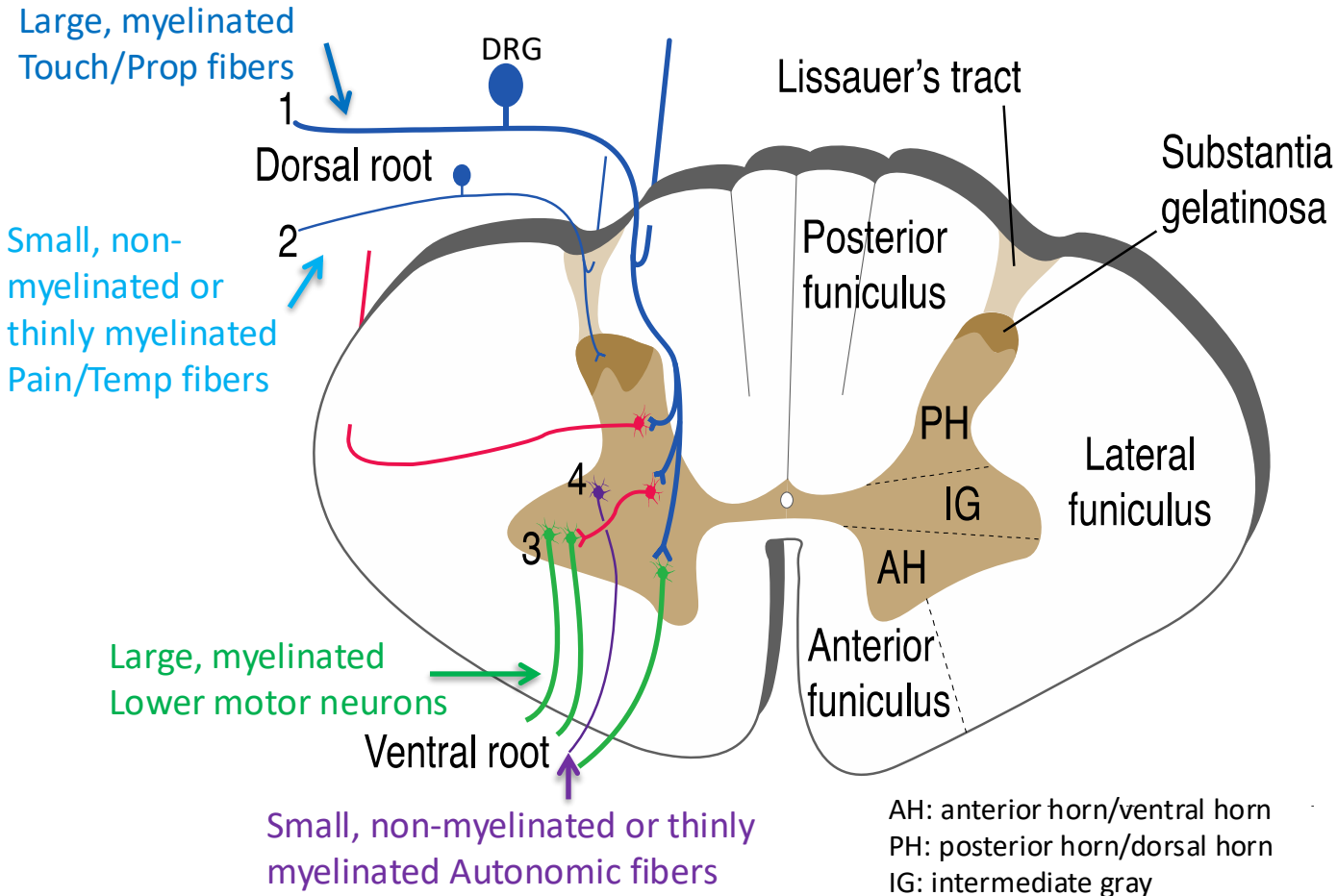
Rash within the dermatome

Usually one side of the body

General Cross-Sectional Anatomy of the Spinal Cord

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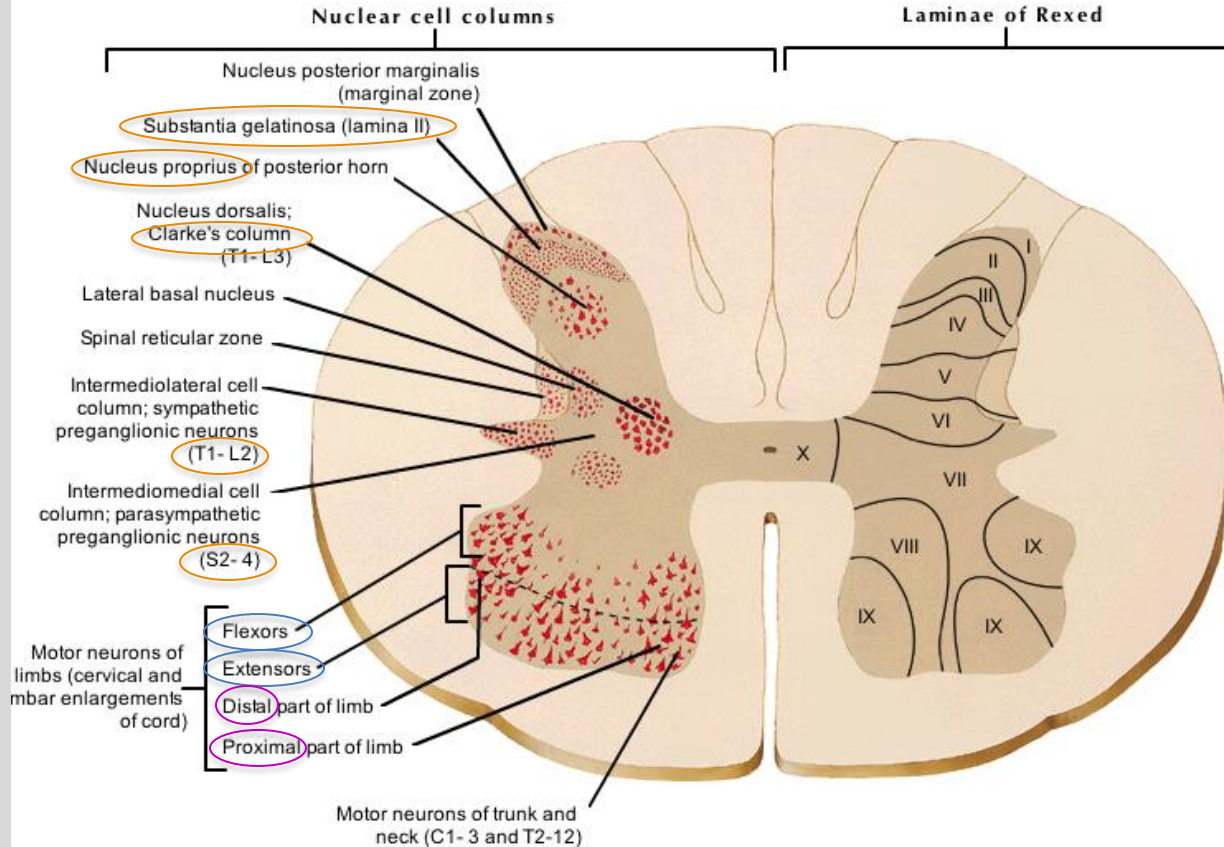
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Laminae of Rexed

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Intrinsic neurotransmitters of the spinal cord

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Excitatory:

- Glutamate (**Glu**): primary afferents, descending projection neurons
- Acetylcholine (**ACh**):
 - autonomic preganglionic neurons
 - lower motor neurons

Inhibitory:

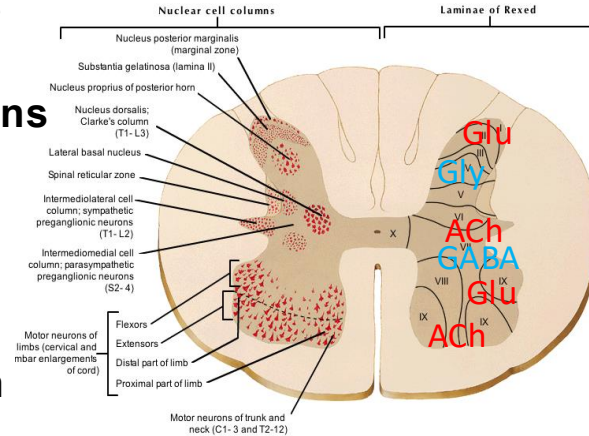
Interneurons (Renshaw cells)

- **GABA**
- Glycine (**Gly**): spinal cord and brainstem only

Neuromodulatory:

primary afferents, descending projection neurons

- Substance P
- Enkephalins
- Calcitonin gene-related peptide (CGRP)



Tetanus

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Cause:

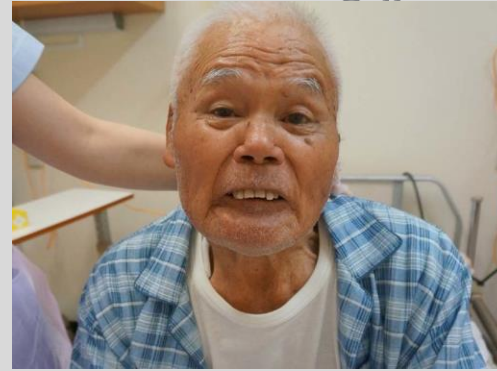
- Clostridia tetani, G+ rod (spores)
 - Infection through wounds
- **Tetanospasmin** (neurotoxin, proteases)
 - Cleaves **SNARE** protein to **block** neurotransmitter release from **inhibitory interneurons (GABA/Gly)**

Symptoms – **painful spastic paralysis:**

- **Brainstem (GABA): risus sardonicus** (sardonic smile), **trismus** (locked jaw) – facial muscle, raised eyebrow and open grin
- **Spinal cord: blocks Renshaw cells (Gly), Opisthotonos** – spinal extensors, most severe form
- **Autonomics:** tachycardia, irritation, restless

Treatment:

- Prevention with vaccine (**10 year** boosters)
- Wound debridement
- Antibiotics/antitoxin
- Diazepam



Osteopathic

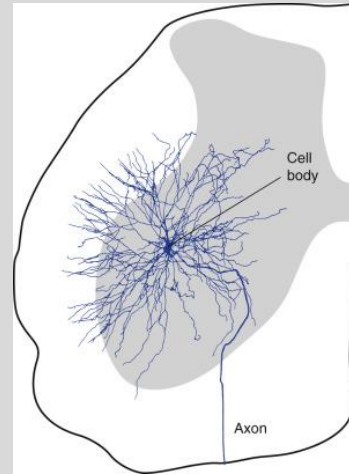


Spinal Reflexes

- Basic patterns of motor coordination
- Regulatory action/negative feedback
- Require no voluntary control or training
- Not sufficient for normal motor performance
- Can be **modified by descending input**

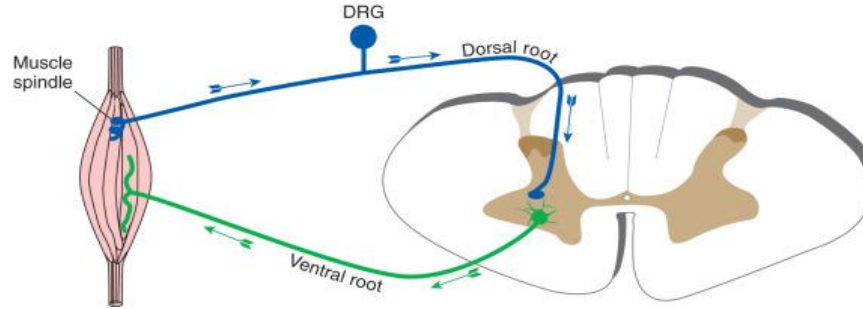


α -motor neuron



Simple Stretch Reflex

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From Nolte J: essentials of human brain, 2nd edition, Fig. 10.3

Sequence

1. Hammer strike **stretches** muscle
2. Muscle spindle is excited
3. Sensory information is conveyed to the spinal cord
4. Excites motor neurons
5. Muscle contraction

- The simplest reflex – **monosynaptic** (one synapse): one primary afferent and one LMN
- Neurotransmitters released:
Glutamate (Glu), Acetylcholine (ACh)
- Also called **deep tendon reflex** (initiated by tapping the tendon)
- Important in posture and locomotion

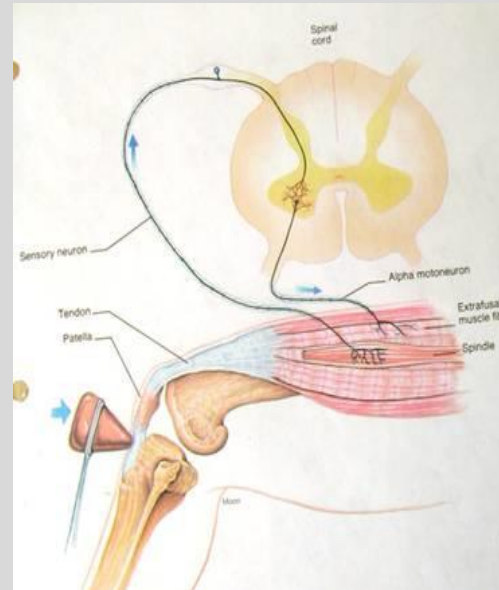
Spinal Distribution of Stretch Reflexes

- Monosynaptic **stretch reflexes** are used in testing the integrity of segments of the cord and the spinal nerves

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Reflex	Cord Segments
Triceps reflex (triceps jerk)	(C6) C7/C8
Biceps reflex (biceps jerk)	C5/C6
Brachioradialis reflex (supinator jerk)	C5/C6
Quadriceps/Patellar reflex (knee jerk)	(L2) L3/L4
Gastrocnemius/Achilles reflex (ankle jerk)	S1/S2



First Aid **mnemonics**:

1/2 “buckle my shoe” -
Achilles reflex (S1, S2)

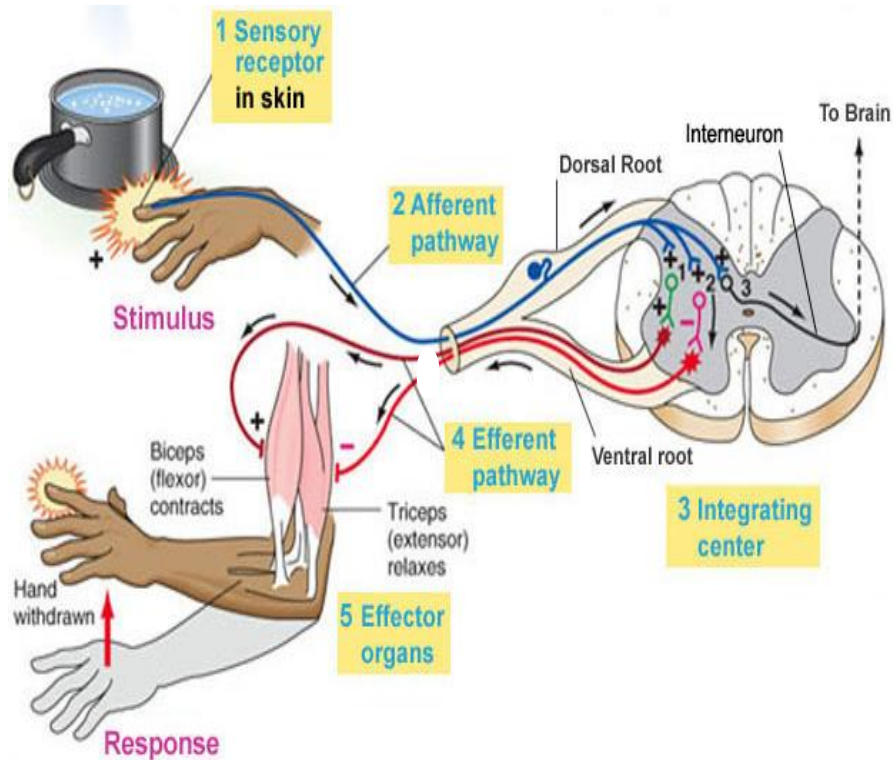
3/4 “kick the door” -
Patellar reflex (L3, L4)

5/6 “pick up sticks” -
Biceps and
brachioradialis reflexes
(C5, C6)

7/8 “lay them straight” -
Triceps reflex (C7, C8)

Flexor Withdrawal Reflex

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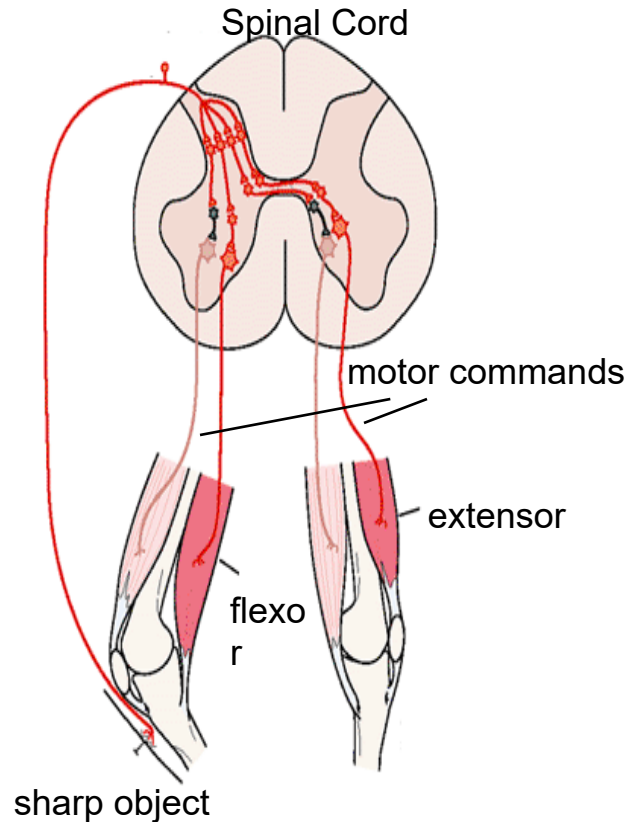


- More complex reflex
 - **Interneurons**
 - Multiple muscles
- Neurotransmitters:
Glu, ACh, Gly

Sequence

1. Painful stimulus on hand
2. Pain sensor is excited
3. Sensory information is conveyed to the spinal cord
4. Usually pass the information to the interneurons to
 - 1) excite flexor motor neurons of the arm, and
 - 2) inhibit extensor motor neurons of the arm, withdrawing the limb from stimulus
5. The sensory neuron could also synapse with the 2nd order neurons to transmit pain signal

Crossed-extensor reflex



Sequence

1. Painful stimulus on right leg
2. Pain sensor is excited
3. Sensory information is conveyed to the spinal cord
4. Excites flexor motor neurons of the right leg, withdrawing limb from stimulus
5. Excites extensor motor neurons of the left leg to support shift in body weight

Feedback control

- Critical role of **interneurons**
- Complex intrasegmental & intersegmental 'reflex chains'

Spinal White Matter

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Take a deep breath

A coffee break

A short walk

.....



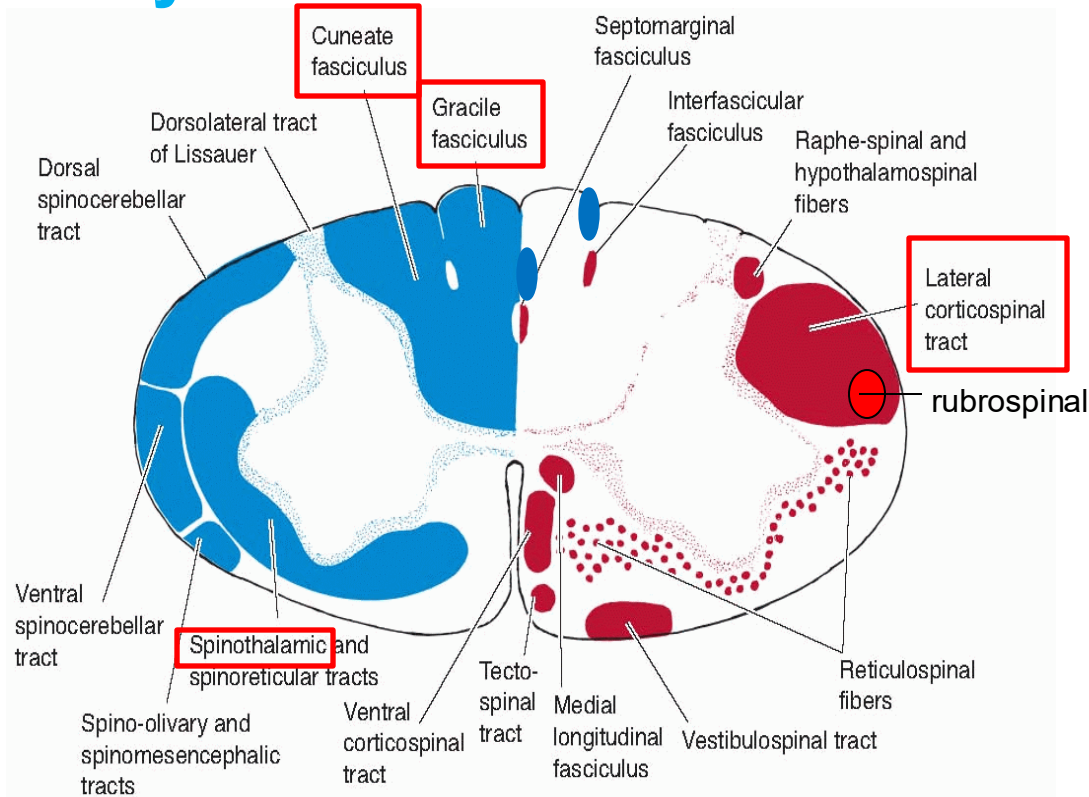
Tracts of the Spinal Cord

Sensory

Motor

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3 major long tracts:

Ascending (sensory):

peripheral → central

- DCML (dorsal column medial lemniscus)
- STT (spinothalamic tract)

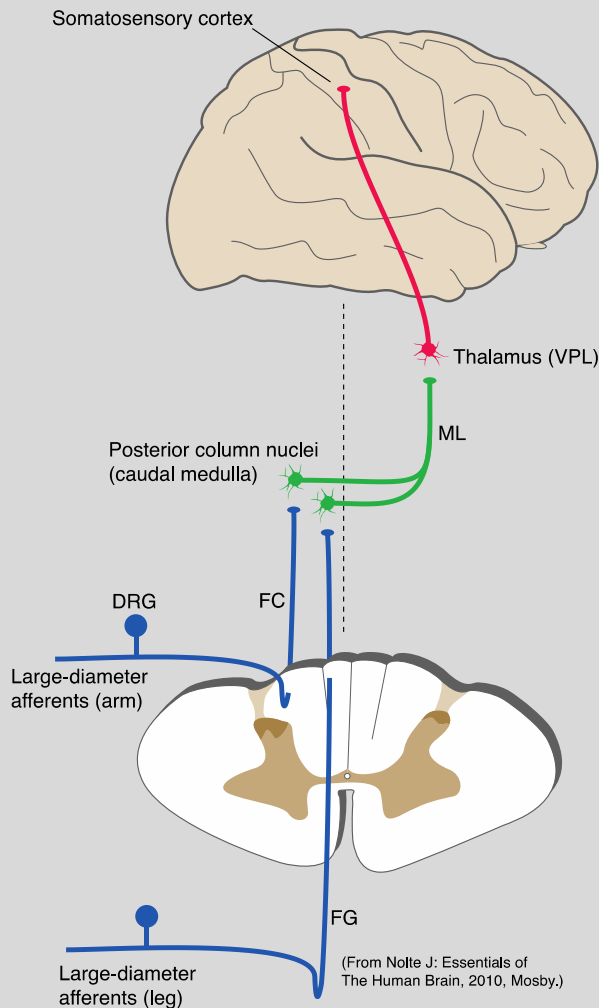
Descending (motor):

central → peripheral

- CST (corticspinal tract)

Ascending Pathway for Touch and Proprioception Sensation DCML Pathway

- **Large diameter afferents** that transmit touch and proprioception enter the dorsal horn
- Touch **1° afferents** travel up the **posterior funiculus (aka posterior column, dorsal column)** **ipsilaterally** to the medulla
- Touch/Prop. from **lower limbs/body** travel in the **Fasciculus Gracilis** (FG)
- Touch/Prop. from **upper limbs/body** travel in the **Fasciculus Cuneatus** (FC)



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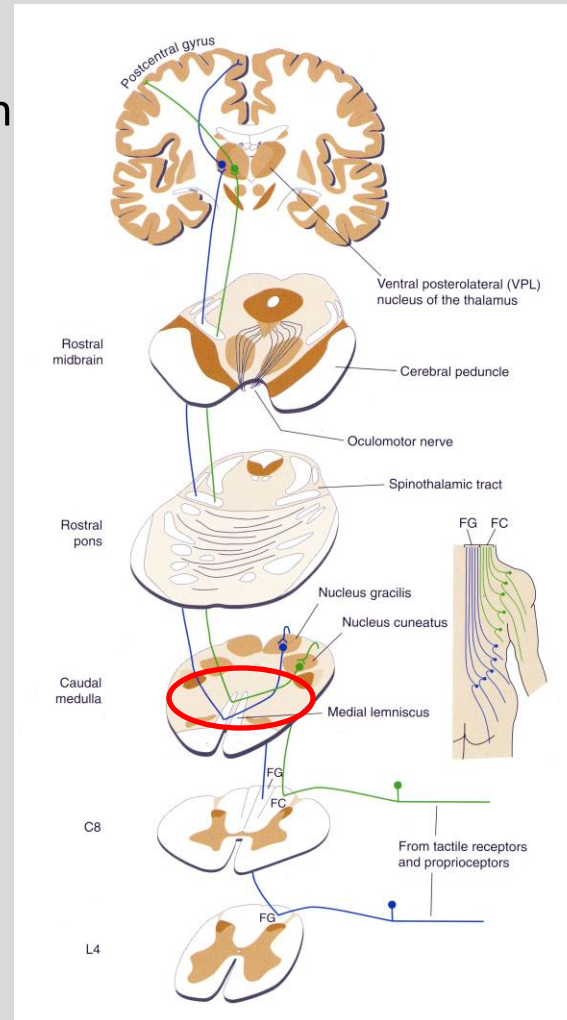
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(From Nolte J: Essentials of
The Human Brain, 2010, Mosby.)

Ascending Pathway for Touch and Proprioception Sensation DCML Pathway

- **Third order neurons** of the **thalamus** take this information to the **somatosensory cortex**
- **Decussation**: These 2nd order neurons cross the midline and join the **contralateral medial lemniscus (ML)** tract on their way to the thalamus.
- **Secondary order neurons'** cell bodies for touch and proprioception are in the **nucleus gracilis (leg)** and **nucleus cuneatus (arm)** of the **medulla**.

From Nolte J: essentials of human brain, 2nd edition, Fig. 10.5



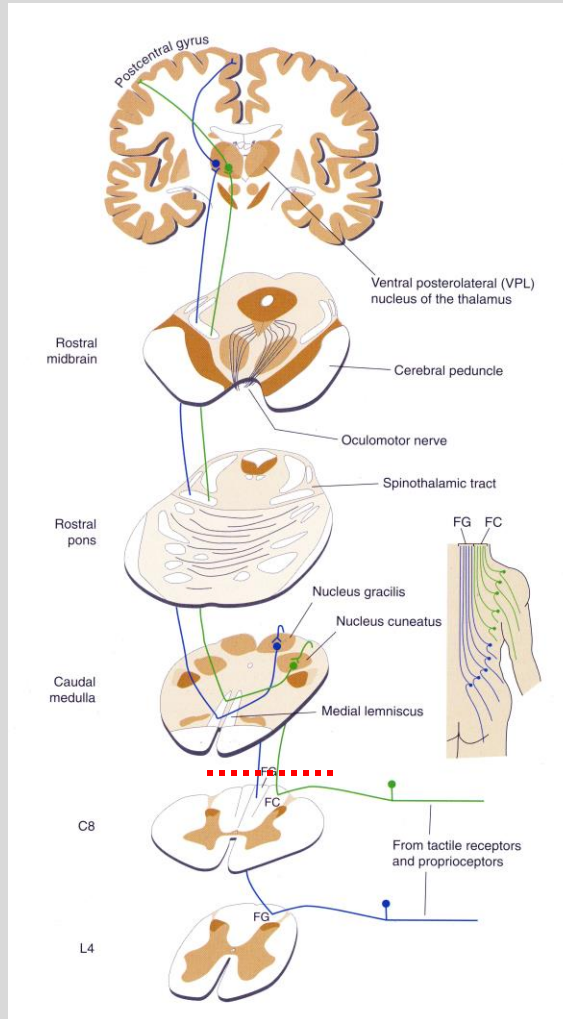
Ascending Pathway for Touch and Proprioception Sensation DCML Pathway

For ascending tracts:

Lesions **before** the
decussation level cause
ipsilateral deficits.

Lesions **after** the
decussation level cause
contralateral deficits.

From Nolte J: essentials of human brain, 2nd edition, Fig. 10.5



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Ipsilateral:

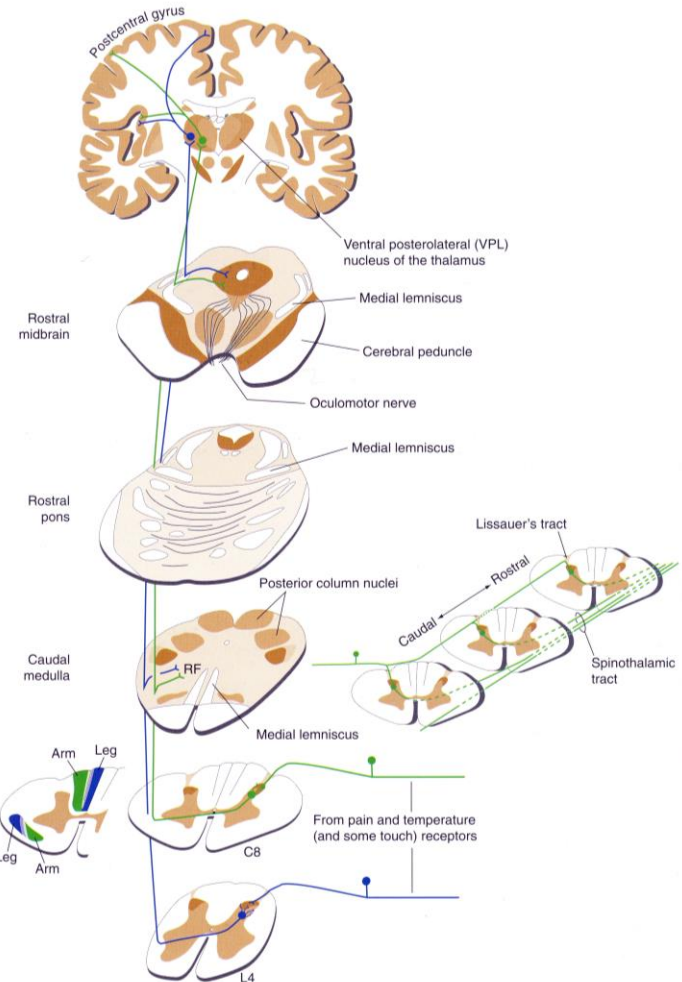
The lesions and deficits
are on the **same side**

Contralateral:

The lesions and deficits
are on the **opposite side**

Ascending Pathway for Pain and Temperature Sensation Spinothalamic Tract

- Pain signals come into the cord from **small diameter primary afferents** and diverge into a spinal level above and a spinal level below in a pathway called **Lissauer's tract**
- Synapse at the **ipsilateral dorsal horn**
- **Decussation**: 2nd order neuron axons cross the midline at **anterior white commissure**
- **Contralateral** spinothalamic tract (STT) - **leg** being more **lateral** and **arm** more **medial** (opposite of DCML)



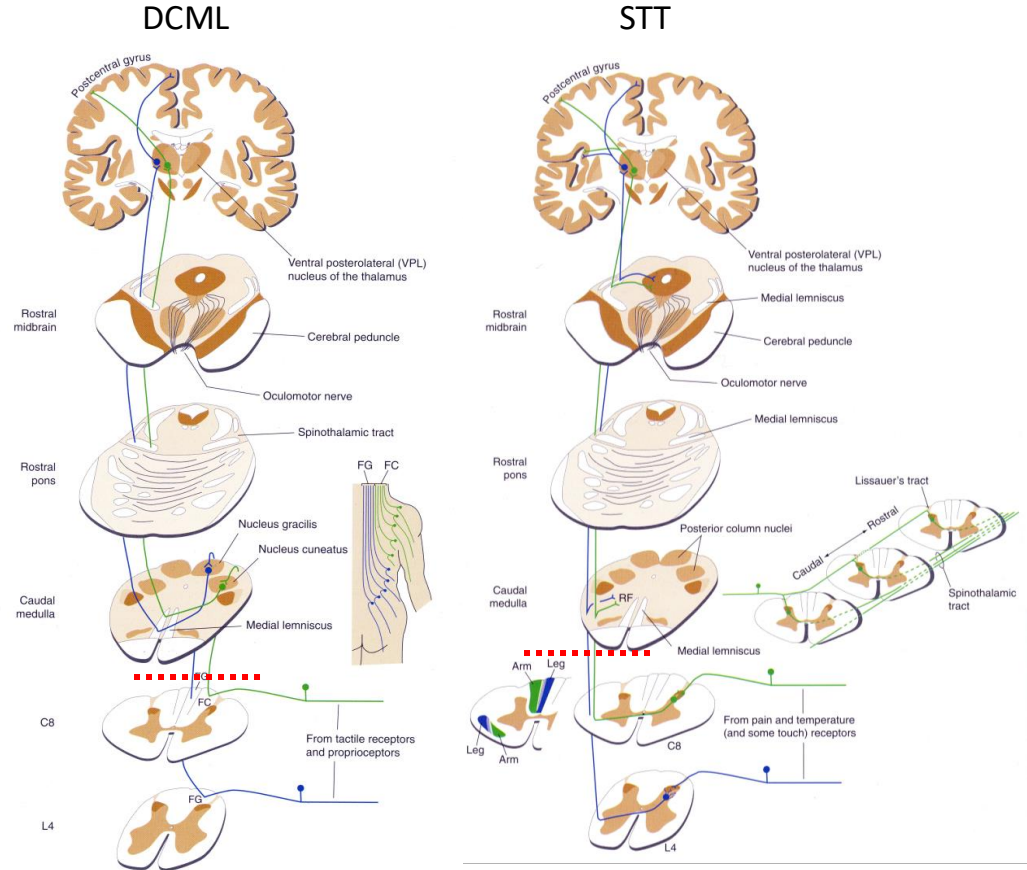
Ascending Pathways

Lesions **before** the decussation level cause **ipsilateral** deficits.

Lesions **after** the decussation level cause **contralateral** deficits.

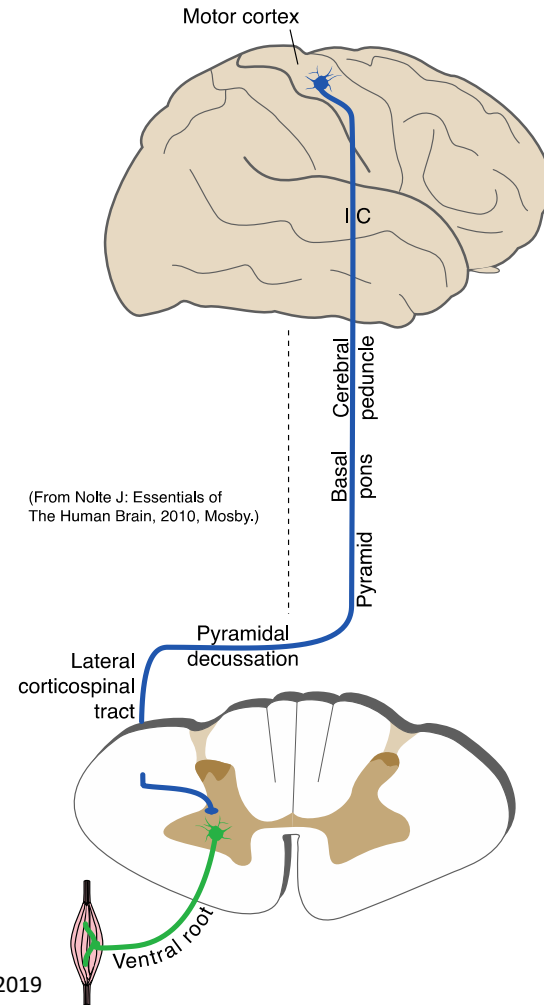
Lesions at the **spinal cord** cause

- **ipsilateral** touch/proprioception
- **contralateral** pain/temperature deficits
- below the level of lesions



Descending Pathway for Somatic **Motor** Activity **Corticospinal Tract**

- **Voluntary movement: upper motor neurons (UMNs)**
- Start in **motor cortex** (*precentral gyrus of the frontal lobe*) and descend through the internal capsule and brainstem
- Most of the fibers cross the midline in the **pyramidal decussation** (at the **spinomedullary junction**) and form the **lateral corticospinal tract** (*lateral funiculus*)
- Synapse at the **contralateral** spinal cord **ventral horn** to activate **lower motor neurons (LMNs)**



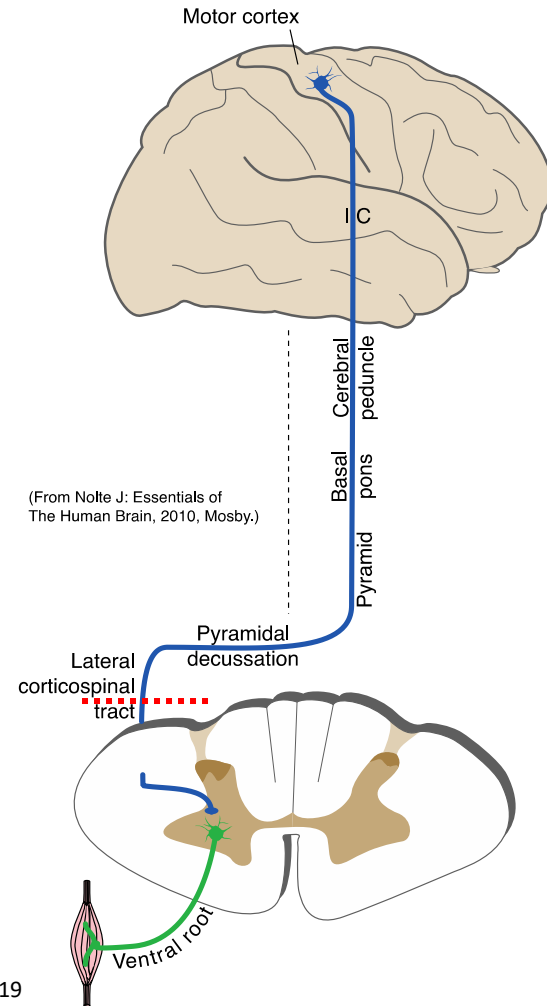
Descending Pathway for Somatic **Motor** Activity Corticospinal Tract

For descending tracts:

Lesions **before** the decussation level cause **contralateral** deficits.

Lesions **after** the decussation level cause **ipsilateral** deficits.

Lesions at the **spinal cord** cause **ipsilateral** paralysis below the level of lesions.



General Rules of Lateralization

For descending (motor) tracts:

Lesions **before** the decussation level cause **contralateral** deficits.

Lesions **after** the decussation level cause **ipsilateral** deficits.

For ascending (sensory) tracts:

Lesions **before** the decussation level cause **ipsilateral** deficits.

Lesions **after** the decussation level cause **contralateral** deficits.

General Rules of Lateralization

- my simplified version

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Lesions **below** the decussation level cause **ipsilateral** deficits.

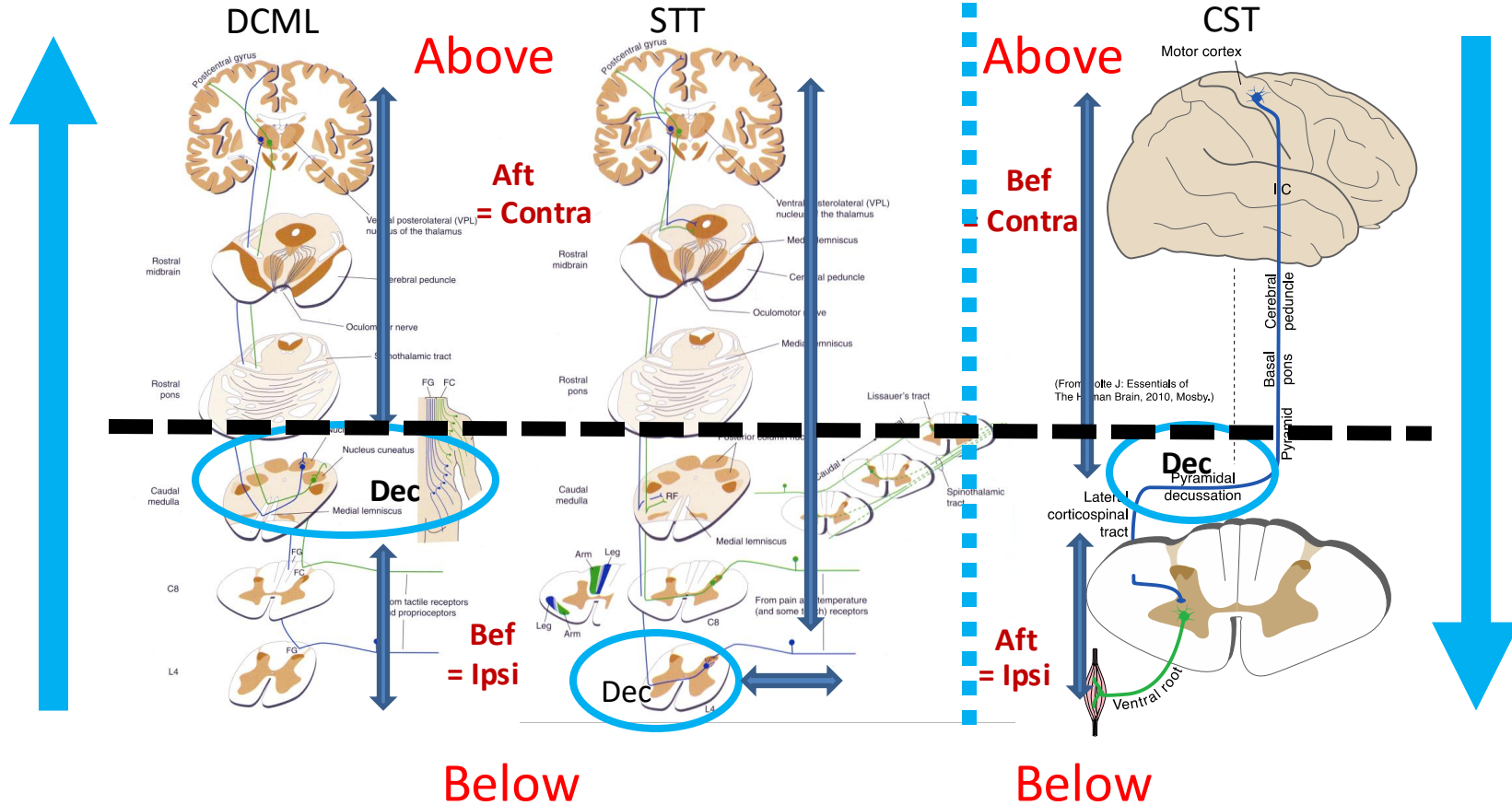
Lesions **above** the decussation level cause **contralateral** deficits.

This is true for **both** ascending and descending tracts.

- For **ascending** tracts, **before** decussation = **below** decussation
- For **descending** tracts, **before** decussation = **above** decussation

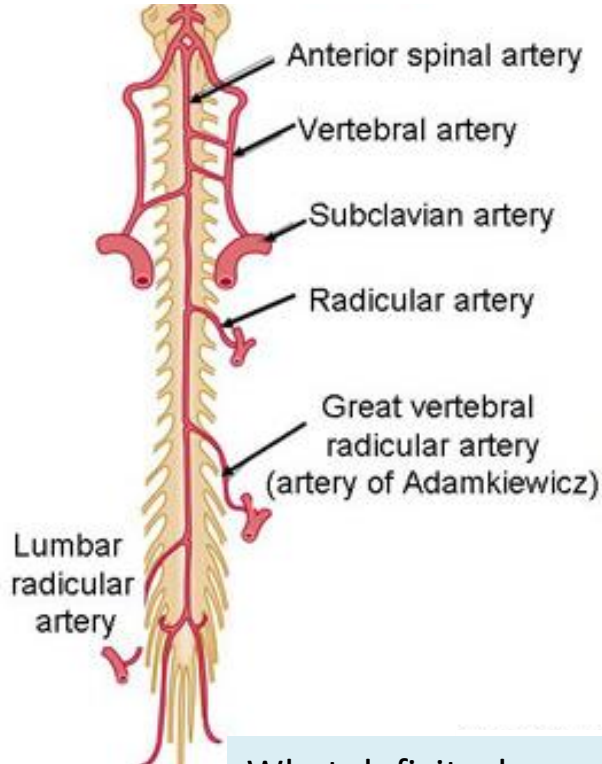
Ascending

Descending

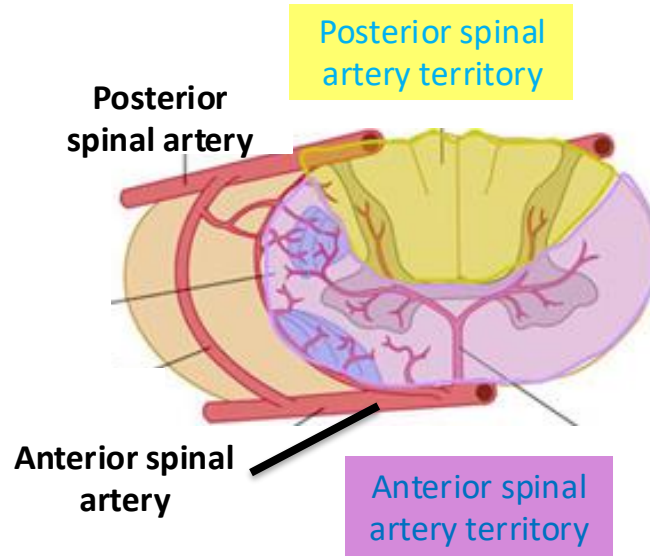


Vascular Supply of the Spinal Cord

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Anastomosis = connection of structures
The posterior spinal arteries have anastomosis connections



What deficits do you expect with **anterior spinal artery stroke**?
What about **posterior spinal artery stroke**?

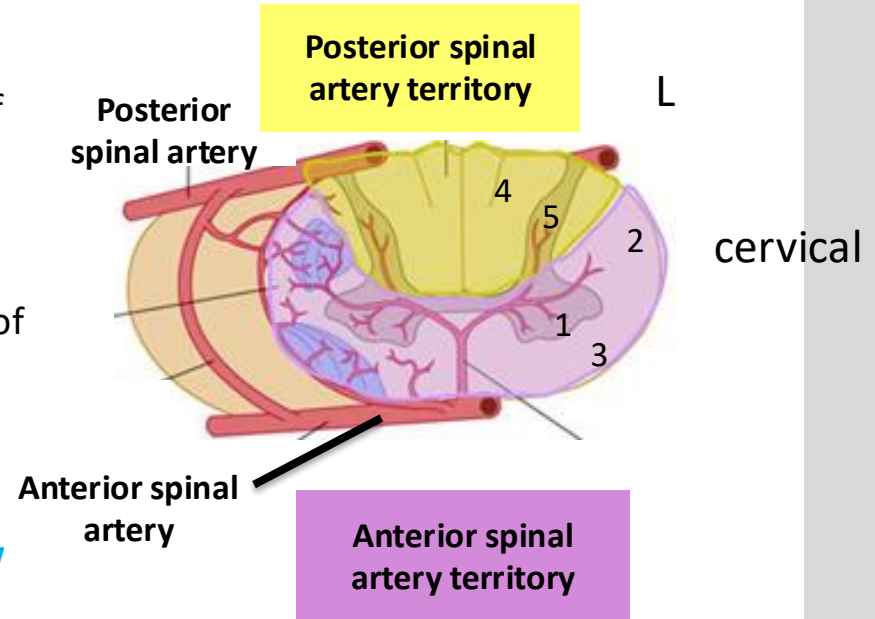
Blood Supply of the Spinal Cord

What deficits do you expect with
anterior spinal artery stroke?

1. LMN deficits (ventral horn) **at** the level of lesion (**ipsilateral, myotome only**)
2. UMN deficits (CST) **below** the level of lesion (**ipsilateral**)
3. Pain/temp deficits (STT) **below** the level of lesion (**contralateral**)

Posterior spinal artery stroke?

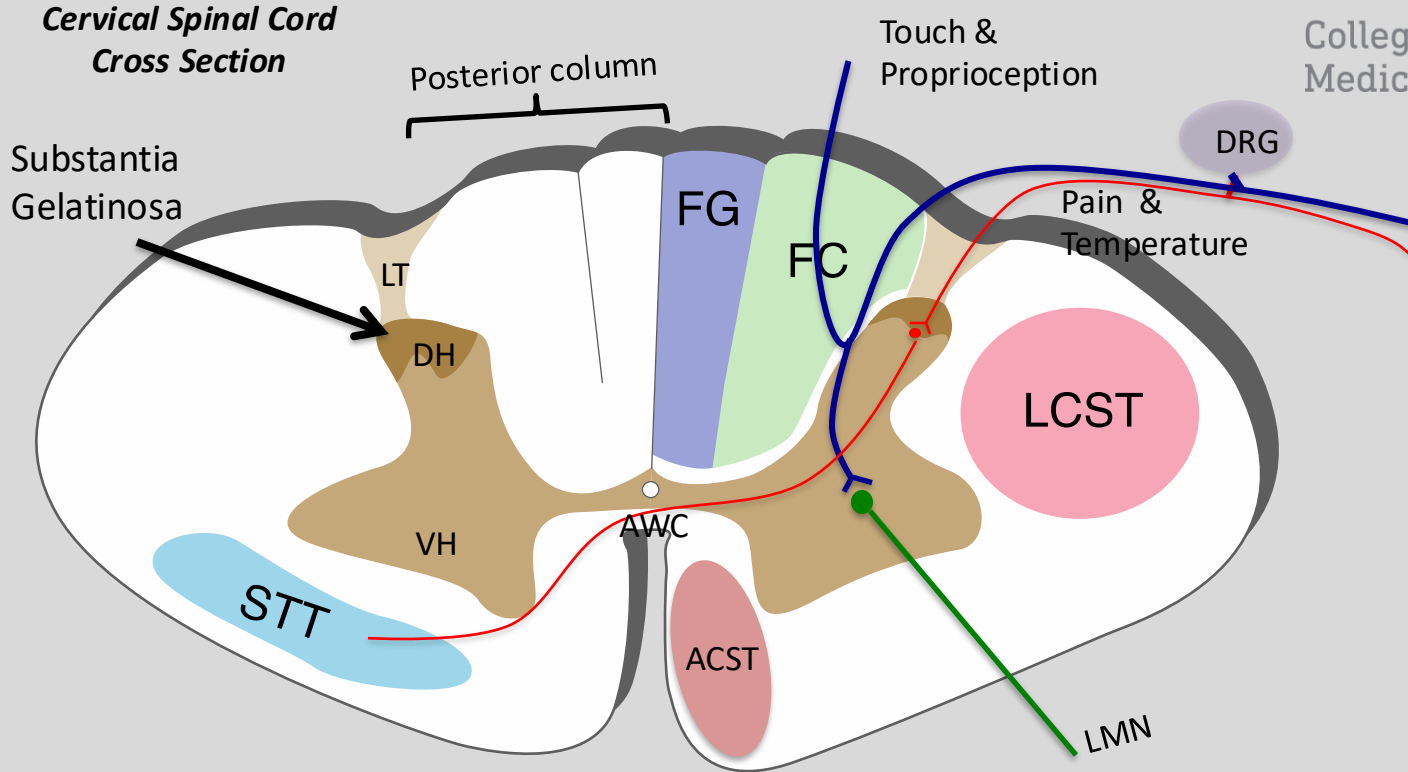
4. Fine touch/proprioception (DCML) **below** the level of lesion (**ipsilateral**)
5. Pain/temp deficits (dorsal horn) **at** the level of lesion (**ipsilateral, dermatome only**)



Summary

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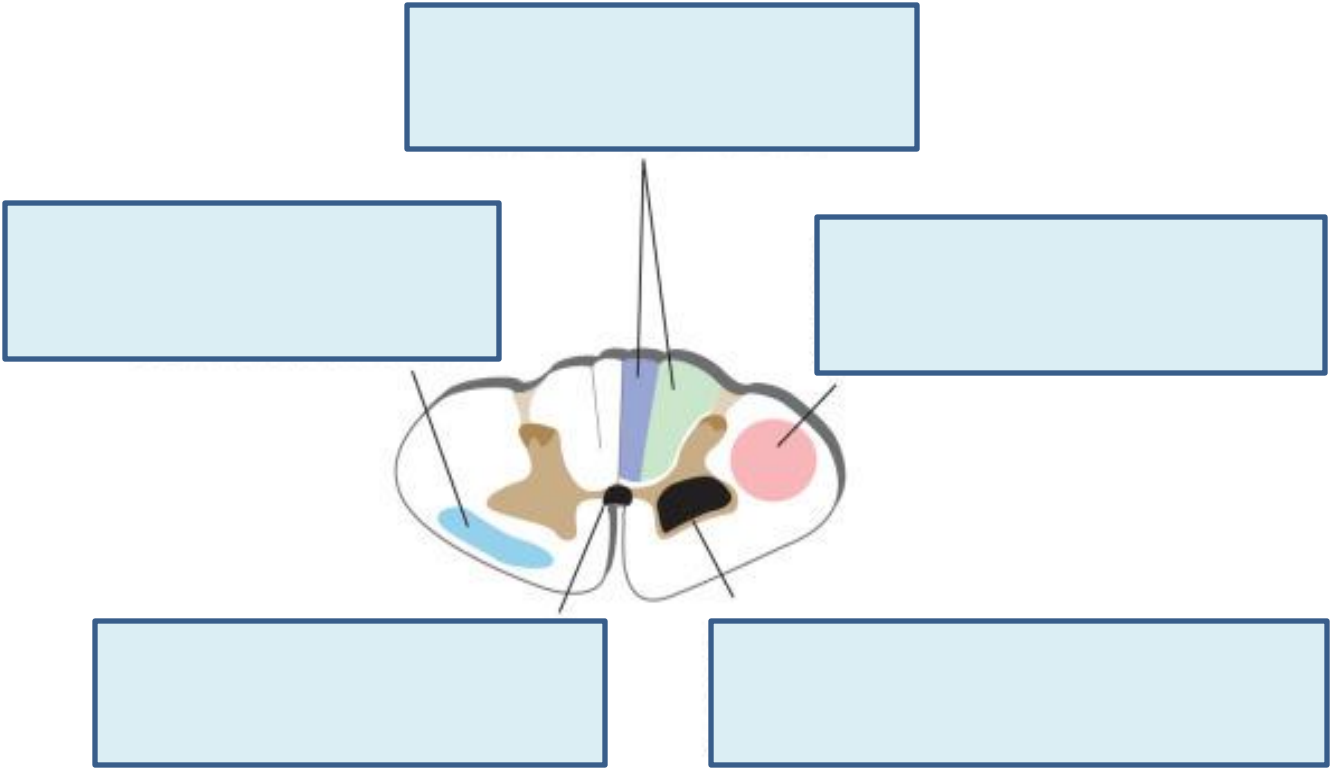


(From Nolte J: Essentials of The Human Brain, 2010, Mosby.)

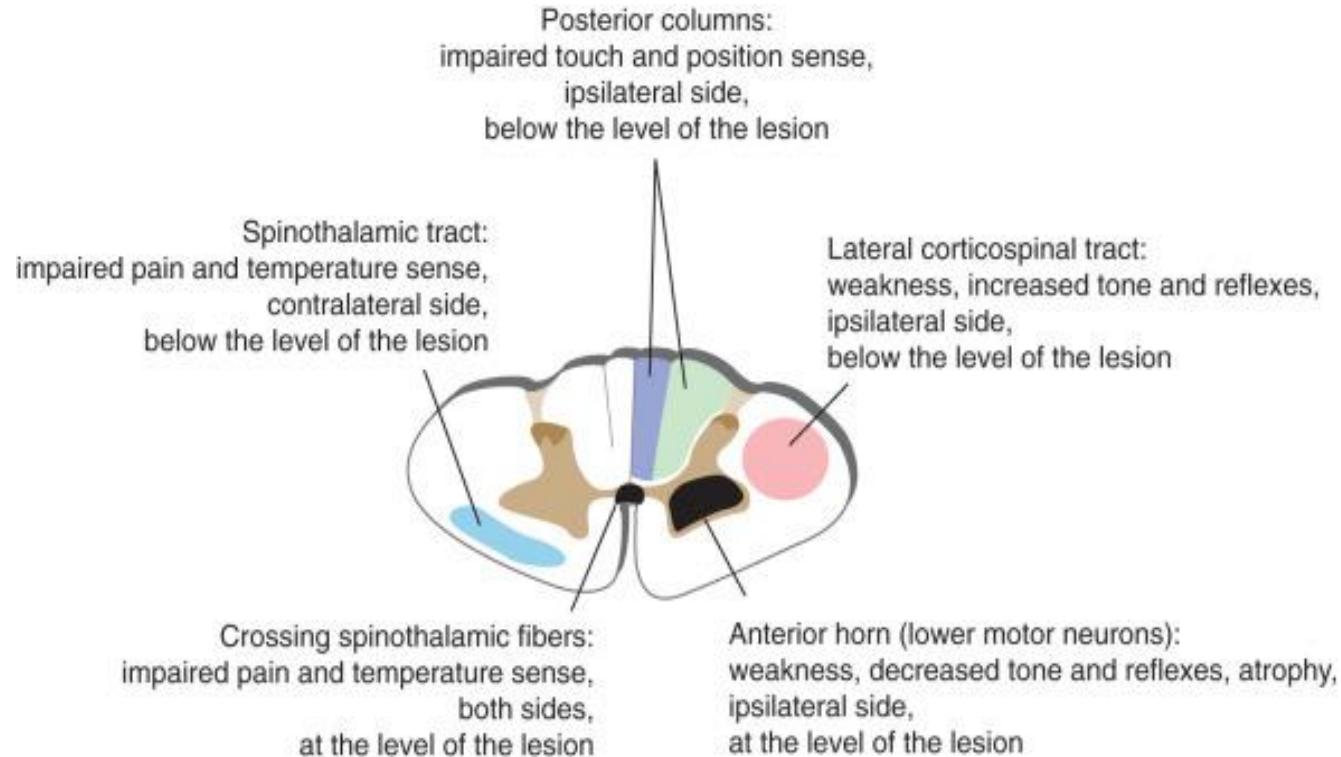
Test yourself

Summary

athic



Summary



Lecture Feedback Form _ Dr. Jennifer Xie

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>



Email: jennifer.xie@nyit.edu